

8TH GRADE MATH • SUPPORT NOTES

Unit 5: The Pythagorean Theorem

NC.8.G.6, NC.8.G.7, NC.8.G.8

Name: _____

Date: _____

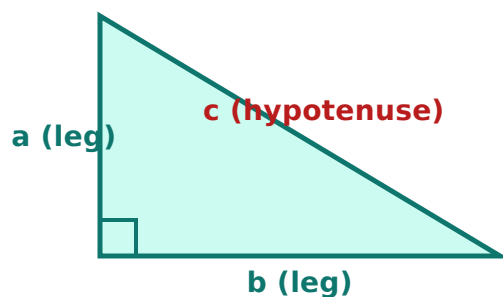
Class: _____

Period: _____

★ The One Big Idea

$$a^2 + b^2 = c^2$$

In a **right triangle**, the two short sides squared and added equal the long side squared.



□ Words to Know

Right triangle — a triangle with one 90° (square corner)

Leg — one of the two shorter sides (a and b)

Hypotenuse — the longest side, always across from the right angle (c)

Square root ($\sqrt{}$) — undoes squaring; $\sqrt{25} = 5$

★ Warm-Up: Squares & Square Roots

📈 Fill in

n	3	4	5	6	8	12	13
n ²							

★ Square Root Undoes Squaring

If $c^2 = 25$, then $c = \sqrt{25} = 5$. The last step is almost always a square root.

5.2 Finding the Hypotenuse (the long side)

NC.8.G.7

 Steps

1. Write $a^2 + b^2 = c^2$.
2. Square the two legs and add.
3. Square root the total to get c.

 I Do: legs 3 and 4

$$3^2 + 4^2 = c^2 \rightarrow 9 + 16 = 25 \rightarrow c = \sqrt{25} = \mathbf{5}$$

 You Try: legs 6 and 8

$$6^2 + 8^2 = \underline{\quad} + \underline{\quad} = \underline{\quad} \rightarrow c = \underline{\quad}$$

5.3 Finding a Missing Leg

NC.8.G.7

 **Different! You must SUBTRACT**

When the missing side is a **leg**, move the known leg over by subtracting.

$$\text{leg}^2 = c^2 - (\text{other leg})^2$$

 **I Do: hypotenuse 13, one leg 5**

$$a^2 + 5^2 = 13^2 \rightarrow a^2 + 25 = 169 \rightarrow a^2 = 144 \rightarrow a = \sqrt{144} = \mathbf{12}$$

 **You Try: hypotenuse 10, one leg 6**

$$a^2 = 10^2 - 6^2 = \underline{\quad} - \underline{\quad} = \underline{\quad} \rightarrow a = \underline{\quad}$$

5.4 Converse: Is It a Right Triangle?

NC.8.G.6

 The Test

1. Find the longest side — that is your possible c .
2. Check: does $a^2 + b^2 = c^2$?
3. **Equal** → YES, right triangle. **Not equal** → not a right triangle.

 I Do: sides 6, 8, 10

$6^2 + 8^2 = 36 + 64 = 100$ and $10^2 = 100$. Equal! → **Yes, right triangle**

5.5 & 5.6

Word Problems & Distance on the Grid

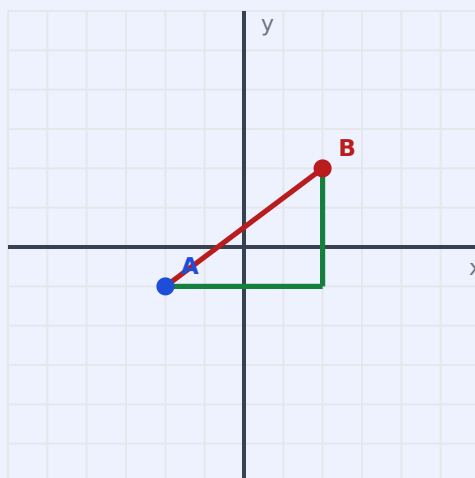
NC.8.G.7, NC.8.G.8

Word Problem Plan

1. Draw the right triangle.
2. Label the two legs and the hypotenuse.
3. Decide: looking for hypotenuse (add) or a leg (subtract)?
4. Solve.

Distance Between Two Points

Make a right triangle: the horizontal change is one leg, the vertical change is the other leg, and the distance is the hypotenuse.



legs 4 and 3 $\rightarrow$ distance = $\sqrt{16+9} = 5$

Part 2 Practice

 Solve (decide: add for hypotenuse, subtract for a leg)

#	Given	Find	Answer
1	legs 9 and 12	hypotenuse	
2	legs 5 and 12	hypotenuse	
3	hyp 15, leg 9	other leg	
4	hyp 17, leg 8	other leg	
5	sides 7, 24, 25	right triangle?	
6	sides 4, 5, 6	right triangle?	

★ ANSWER KEY ★ **Guided Notes**

Warm-up: 9,16,25,36,64,144,169. 5.2 You try: $36+64=100$, $c=10$. 5.3 You try: $100-36=64$, $a=8$.

 Practice

1) 15 2) 13 3) 12 4) 15 5) Yes ($49+576=625=25^2$) 6) No ($16+25=41 \neq 36$)

End of Unit 5 support packet.